GAURAV MITTAL	
CONTACT INFORMATION Email: gaurav198511@gmail.com Phone: +91-9634888123 Address: Bareilly Road, Near Dhan Mill, "Kamal Kunj", Haldwani, Uttarakhand DOB: 19th November 1985	

PROFILE SUMMARY

Accomplished academic professional with 13+ years of comprehensive experience in engineering education, corporate training, and research. Currently pursuing Ph.D. and MTech from prestigious institutions, demonstrating commitment to continuous learning and academic excellence. Proven expertise in delivering high-quality education in emerging technologies, including Artificial Intelligence, Machine Learning, Data Science, and Cloud Computing.

Academic Excellence:

- Extensive Teaching Experience: Successfully taught diverse engineering subjects across multiple institutions with a focus on cutting-edge technologies
- Research Leadership: Led multiple research projects funded by SERB-DST, NIF-DST, and institutional grants with significant societal impact
- Industry-Academia Bridge: A Unique combination of academic rigor and industry experience, bringing real-world applications to classroom learning
- Student Development: Proven track record in student mentoring, placement coordination, and skill development programs
- Innovation in Education: Designed and implemented specialized courses, laboratory setups, and hands-on training modules

Research & Development:

- Active Researcher: Multiple publications in peer-reviewed journals and international conferences
- Grant Writing: Successfully secured research funding from prestigious government agencies
- Technology Transfer: Translated research outcomes into practical applications with industry relevance
- Collaborative Research: Extensive experience in interdisciplinary research projects and industry partnerships

EDUCATION

Degree	Institution	Year
Ph.D. (Pursuing)	GKV (Deemed University), Haridwar	Current
MTech	IIT Patna	2025
Research Fellow	UPES Dehradun	2018-2019
MTech.	MANIT Bhopal	2013
B.Tech.	UPTU University	2009

KEY SKILLS

Technical Skills

- Programming: Python, C, C++, Statistics & Data Cleaning
- AI/ML: Machine Learning, Artificial Intelligence, Deep Learning, Generative AI
- Data Science: Data Analytics, Data Visualization, TensorFlow
- DevOps: Docker, Kubernetes, Jenkins, Terraform
- Cloud Computing: AWS EC2/RDS, Cloud Implementation
- Database: Oracle DBA 19c, PostgreSQL, Oracle Data Guard, Oracle Golden Gate 12c
- Tools: Linux Administration, RMAN Backup & Recovery, Incident Management

Training & Development

- Corporate Training in AI & ML
- Workshop Design and Delivery
- Curriculum Development
- Technical Content Creation
- Student Mentoring and Guidance

WORK EXPERIENCE (12 years 8 months)

1- Assistant Professor | Roorkee Institute of Technology (Aug 2024 - Present)

Department: Computer Science & Engineering & AIML

Roles & Responsibilities:

- DevOps Trainer specializing in modern deployment practices
- AI & ML Trainer with Cloud Implementation expertise
- Placement & Training Coordinator for student career development

- Course curriculum design for emerging technologies

2- Assistant Professor | Tula's Institute, Dehradun (Feb 2023 - Aug 2024)

Department: Computer Science & Engineering

Specialization: Database Systems (SQL & NoSQL) and Data Science Engineering

Key Achievements:

- Coordinated specialization subjects for B.Tech CSE with focus on AI & ML
- Served as VAP Coordinator for B.Tech Computer Science program
- Submitted research projects for external funding to DST and UCOST
- Active member of the Research and Development team

3- Oracle Database Administrator | Impact Infotech (Oct 2022 - Feb 2023)

Responsibilities:

- Managed incidents, changes, and problems effectively
- Conducted daily, weekly, and monthly server maintenance
- Performed Oracle RAC database patching without downtime
- Executed database refreshes using RMAN duplicate command
- Migrated databases to AWS Cloud using Data Pump

4- Oracle Database Administrator Level II | Osho Industries Ltd (Dec 2018 - July 2022)

Key Accomplishments:

- Managed Oracle Database versions 11g, 12c, and 19c
- Upgraded 25+ databases from legacy versions to 19c (on-premises and cloud)
- Handled 25+TR database backup using RMAN on Cohesity disks
- Configured Golden Gate Uni-directional and Bi-directional replication
- Developed shell scripts for ADR and audit log purging

5- Assistant Professor & Skill Trainer | College of Engineering Roorkee (Aug 2013 - Nov 2018)

Major Projects:

- Led a 2-crore automation project at the Centre of Excellence
- Technical Manager for 1.5 crore NABL lab project
- Managed budgeting and procurement for projects totalling 5-crore
- Provided training for lab instructors and assistants

6- Site Engineer | DMRC (Under A to Z Infra Services) (July 2009 - Feb 2011)

- Testing and commissioning at substations
- SCADA and Distributed Control Systems management over TCP/IP

RESEARCH PROJECTS & PUBLICATIONS

Major Research Projects

1. **SERB-DST Funded Project (2018-2021)**

   Title: "Ventilation Augmented Supercavitating Turbine (VAST) Technology for Next-Generation Hydropower"

   - Role: Senior Research Fellow at MPFL Lab, UPES Dehradun
   - Funding Agency: Science and Engineering Research Board, DST India
   - Project Scope: Developed proof of concept for next-generation hydropower technology through various experimental setups

     Key Contributions:

     - Designed and established experimental facilities for turbine efficiency analysis
     - Conducted computational fluid dynamics simulations for turbine performance optimization
     - Developed mathematical models for supercavitating turbine operations
     - Published research findings in international conferences and journals

2. **COVID Innovation Challenge Project (2020)**

   Title: "Design, Fabrication, and Pilot Study of Comfortably Vented Indigenously Designed (COVID) Fabric Helmet"

   - Role: Co-Principal Investigator
   - Funding Agency: National Innovation Foundation, DST India
   - Innovation Impact: Developed breakthrough personal protective equipment during pandemic
   - Media Recognition: Featured in major news outlets including Times of India, Indian Express, and Red FM
   - Social Impact: Addressed critical healthcare safety needs during COVID-19 pandemic

3. **Hybrid Energy Vehicle Development Project**

   Title: "Modelling & Development of Laboratory Model for Hybrid Energy Vehicle"

- Project Scope: Integration of multiple energy resources (Solar, Battery, Ultracapacitors)
- Technical Achievement: Developed DC-DC converter-based power train system with regenerative braking
- Innovation: Utilized BLDC motors for traction with efficiency optimization algorithms
- Outcome: Successfully demonstrated hybrid energy integration for sustainable transportation

4. **UPES SEED Grant Projects (2018)**
    - Project 1: "Novel strategies to induce bubble coalescence for industrial and naval applications"
    - Project 2: "In-situ monitoring of industrial bubbly flows and Bubble size control by continuous feedback system"
    - Project 3: "Development of test facility for efficiency measurement of novel hydrokinetic turbine designs"
5. **Toycathon-2021 National Competition**
    - Achievement: Winner in Digital Toycathon category for "Mission Setubandha" game
    - Recognition: Team interaction with Prime Minister Narendra Modi before Grand Finale
    - Innovation: Promoted Arogya and Ayurveda through innovative educational board game

Ph.D. Research (Ongoing)

- University: GKV (Deemed University), Haridwar

M.Tech Research Projects

- Current Project (IIT Patna): Cloud Computing architecture and optimization
- Completed Project (MANIT Bhopal): "Modeling and simulation of grid-connected inverter for SPV on Matlab"

TRAINING ASSIGNMENT AT IITS

Comprehensive Workshop Series at Premier Indian Institutes of Technology

2025 Training Programs:

- **IIT Hyderabad:**
    - AI & ML Masterclass (June 14-15, 2025) - Advanced machine learning techniques and applications in collaboration with EduNautic's and Tinkerers' Lab

    - Data Science with Visualization Using Python (June 10-12, 2025) - Comprehensive data analytics and visualization workshop with EduNautic's and Tinkerers' Lab
    - Generative AI Workshop (March 8, 2025) - Cutting-edge generative AI technologies with World Technocon
    - Machine Learning & Data Analytics (March 9, 2025) - Advanced ML algorithms and data processing with World Technocon
- **IIT Delhi:**
    - AIML Workshop (May 17, 2025) - Artificial Intelligence and Machine Learning fundamentals with World Technocon
    - AIML Workshop (May 11, 2025) - Advanced AI/ML applications with World Technocon
    - Generative AI Workshop (March 2, 2025) - Latest developments in generative AI with World Technocon
    - Machine Learning & Data Analytics (March 1, 2025) - Comprehensive ML and analytics training with World Technocon
- **IIT Roorkee:**
    - Generative AI Workshop (February 2, 2025) - Advanced generative AI techniques with World Technocon
    - Machine Learning & Data Analytics (February 1, 2025) ML algorithms and data analysis with World Technocon
    - Data Analytics Workshop (January 25-26, 2025) - Comprehensive data analytics training with Techgyan Technologies

2024 Training Programs:

- AI, ML & Deep Learning - IIT Hyderabad (November 16-17, 2024) - Advanced deep learning concepts with Tinkerers' Lab
- AI, ML & Deep Learning - IIT Bombay, Aavhan 2024 (October 19-20, 2024) Comprehensive AI/ML training with Ethical Edufabrica
- Data Science Workshop - IIT Roorkee (September 21-22, 2024) - Data science methodologies with Remark Skill Education

2023 Training Programs:

- Artificial Intelligence - IIT Indore (September 9-10, 2023) - AI fundamentals and applications with Mun club
- Artificial Intelligence - IIT Roorkee (September 2-3, 2023) - Advanced AI concepts with e-cell

- AI & Machine Learning - IIT Patna (March 25-26, 2023) - Comprehensive AI/ML training with e-cell
- Data Science - IIT Roorkee (January 7-8, 2023) - Data science and analytics with Thomso

Guest Speaking Engagements:

- Guest Speaker - Guru Jambheshwar University of Science and Technology, Hisar (March 23-24, 2025) - Workshop on AI, ML & Data Science

Training Impact & Recognition:

- 19+ Workshop Sessions conducted at premier IITs across India
- 1000+ Students Trained in cutting-edge technologies
- Multi-institutional Collaboration with leading technology clubs and organizations
- Industry Partnership through various EdTech companies for practical training delivery
- Consistent Recognition for innovative training methodologies and content delivery

PUBLICATIONS & RESEARCH OUTPUT

Peer-Reviewed Journal Publications

1. Mittal, G., Donga, R. K., Kumar, A., Karn, A. (2019). "Identification of Silt Erosion Areas on Francis Turbine Guide Vanes Through Numerical Simulation." 46th National Conference on Fluid Mechanics and Fluid Power, PSG Institute of Technology, Coimbatore, India.
2. Sawant, G., Gaurav, K., Mittal, G., Karn, A. (2020). "A short review on the development of novel face masks during COVID-19 pandemic." Proceedings of International Conference on Advances in the Field of Health, Safety, Fire, Environment & Allied Sciences (HSFEA 2020).
3. Gaurav, K., Mittal, G., Karn, A. (2021). "On the Morphology of Elongated bubbles during their formation at submerged orifices." Chemical Engineering Science.

Conference Publications

4. Mittal, G. (2015). "Simulation & modelling of grid-connected solar PV." National Conference NCRIET-2015, COER, Roorkee, March 28-29, 2015.
5. Mittal, G. (2014). "Two-Dimensional Sun Tracker with the Application of Neural Network & Embedded System." National Conference NCETEST-2014, COER, Roorkee, March 29, 2014.
6. Mittal, G. (2014). "Slip Power Recovery Controlled Induction Motor Drive with Fuzzy Logic Controller." National Conference NCETEST-2014, COER, Roorkee, March 29-30, 2014.

Research Impact & Recognition

Media Coverage of Research Work:

- Featured in Times of India (Education Times) for COVID Ventilator development
- Covered by Indian Express (Edex Live) for innovative COVID protective equipment
- Recognized by Rajya Sameeksha for breakthrough healthcare innovations
- Featured on Red FM's "Ek Pahadi Aisa Bhi" program as Corona Hero

Technical Innovation Patents & IP:

- Developed proprietary shell scripts for database maintenance and optimization
- Created innovative educational modules for industrial automation
- Designed novel approaches for hybrid energy system integration

Research Collaboration:

- Collaborated with leading technology companies, including KVTech Gurugram, NEO Engineering Pune
- Industry partnerships for laboratory development and testing facilities
- International collaboration through conference presentations and publications

CORPORATE TRAINING EXPERIENCE

Associated EdTech Companies:

1. EduNautic's - Gurugram, Haryana
2. Techgyan Technologies - Delhi
3. Remark Skill - New Delhi
4. ACODS - Noida, Uttar Pradesh
5. Ethical Edufabrica Pvt. Ltd New Delhi
6. World Technocon - Noida, Uttar Pradesh

RECENT ACTIVITIES & IIT WORKSHOPS

2025 Workshops:

- **IIT Hyderabad Workshops:**
  - AI & ML Masterclass (June 14-15, 2025) - EduNautic's collaboration with Tinkerers' Lab
  - Data Science with Visualization Using Python (June 10-12, 2025) - EduNautic's collaboration with Tinkerers' Lab
- **IIT Delhi Workshops:**
  - AIML Workshop (May 17, 2025) - World Technocon collaboration
  - AIML Workshop (May 11, 2025) - World Technocon collaboration

    - Generative AI Workshop (March 2, 2025) - World Technocon collaboration
    - Machine Learning & Data Analytics (March 1, 2025) - World Technocon collaboration.

- **Other Notable Workshops 2025:**
    - Guest Speaker - Guru Jambheshwar University of Science and Technology, Hisar (March 23-24, 2025) - AI, ML & Data Science
    - Generative AI Workshop - IIT Hyderabad (March 8, 2025) - World Technocon collaboration
    - Machine Learning & Data Analytics - IIT Hyderabad (March 9, 2025) - World Technocon collaboration
    - Generative AI Workshop - IIT Roorkee (February 2, 2025) - World Technocon collaboration
    - Machine Learning & Data Analytics - IIT Roorkee (February 1, 2025) - World Technocon collaboration
    - Data Analytics Workshop - IIT Roorkee (January 25-26, 2025) - Techgyan collaboration

2024 Workshops:

- AI, ML & Deep Learning - IIT Hyderabad (November 16-17, 2024) - Tinkerers' Lab collaboration
- AI, ML & Deep Learning - IIT Bombay, Aavhan 2024 (October 19-20, 2024) - Ethical Edufabrica
- Data Science Workshop - IIT Roorkee (September 21-22, 2024) - Remark Skill Education

2023 Workshops:

- Artificial Intelligence - IIT Indore (September 9-10, 2023) - Mun club collaboration
- Artificial Intelligence - IIT Roorkee (September 2-3, 2023) e-cell collaboration
- AI & Machine Learning - IIT Patna (March 25-26, 2023) e-cell collaboration
- Data Science - IIT Roorkee (January 7-8, 2023) - Thomso collaboration

ONLINE TRAINING PROGRAMS

Recent Internship Programs:

- 4-Week AI & Data Science Internship (June 6 - July 5, 2025) Ethical Edufabrica
- 45-Day AI & Data Science Internship (August 3 - September 20, 2023) - Ethical Edufabrica

- 45-Day ML & AI Internship (September 10 - October 30, 2023) - Ethical Edufabrica

CERTIFICATIONS & TRAINING

- Oracle Database Administration (11g, 12c, 19c)
- Golden Gate Replication Specialist
- AWS Cloud Database Migration
- Python Programming & Machine Learning
- DevOps Tools: Docker, Kubernetes, Jenkins

LANGUAGES

- Hindi (Native)
- English (Fluent)

DECLARATION

I hereby declare that all the information provided above is true and accurate to the best of my knowledge and belief.

Date: July 23, 2025

Place: Roorkee, Uttarakhand

Gaurav Mittal